

Week 9-12-MME 305 Course Project: Pancake Process Improvement (Perfect Precision Pancakes)

Background and Application

You and your groupmate(s) have been hired by a breakfast restaurant chain to improve the quality and consistency of its menu items. Based on the results of a customer satisfaction survey, customers have complained that certain locations produce high-quality meals while other locations are more efficient at delivering meals. To help the restaurant chain improve upon the speed of delivery and improve quality, you and your groupmates will review and improve the process to make the restaurant's chain's classic breakfast staple: Swedish Pancakes. The recipe is included below:

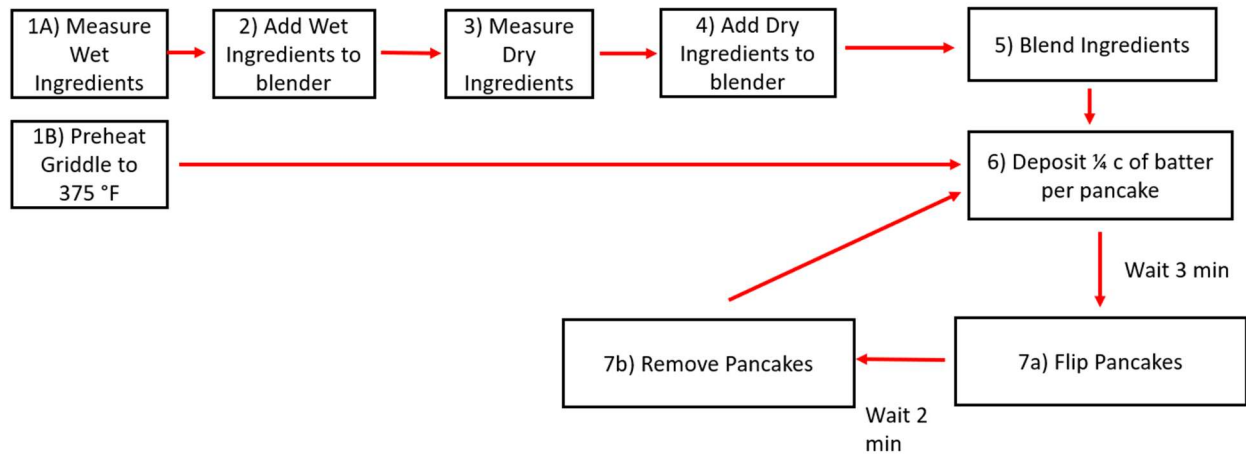
Swedish Pancake Recipe (adapted from: <https://artofnaturalliving.com/desperate-mom-dinners-healthier-swedish-pancakes/>)

- Blend the following in a blender:
 - 1/2 cup milk
 - 1 large egg
 - 1/2 cup all-purpose flour
 - 1 tsp white sugar
 - 1 tablespoon of melted salted butter (**add last so it doesn't solidify**), may use vegetable or canola oil instead of butter as an alternative
- Pour out pancake batter in ~1/4 cup amounts onto a skillet set to 375 °F.
 - Cook on first side for 3 minutes, flip with spatula.
 - Cook on second side for 2 minutes, remove pancake from skillet.
 - Serve with butter, maple syrup, or fresh fruit

Some preliminary mass measurements of these ingredients have been taken and are available in a separate excel file that has been uploaded to CANVAS. Mass measurements are preferred over volumetric measurements in cooking due to measurement consistency. Additionally, your group will add a thermocouple sensor and solid-state relay to monitor and control the temperature of the griddle during the cooking process. This will ensure the griddle temperature remains within an acceptable range during cooking.

After your group has completed the review of the manufacturing process, the group will design a MATLAB app that will be used by a restaurant employee to prepare a batch of pancakes. The app must be easy to use, provide clear instructions to the operator on what to do, and store critical information (mass of each ingredient used and temperature during the cooking process) in a separate data file after the operator has finished cooking. To create the app, you will use MATLAB app designer [1]. This problem models the requirements of many "Industry 4.0" manufacturing processes that allow machine operators to quickly assemble new products quickly with little to no training requirements [2-4].

Pancake Manufacturing Process



Purpose of the Project

The purpose of the project is for your group to develop independent skills in setting up a measurement and data acquisition system to improve a process. There will be no step-by-step procedure on how-to set up the measurement system. It is expected that during the semester you have been able to acquire enough knowledge and skills to complete the project. Please refer to prior laboratory handouts, class notes, and presentation files to help you recall applicable information and use it towards completing the project.

After completing the project, the student should learn the following skills:

1. Use statistical analysis to identify and eliminate outliers in a data set (see class notes on statistics posted on CANVAS).
2. Design a measurement and control system that includes the following components.
 - a. Load Cell & Load Cell Amplifier
 - b. Thermocouple and Thermocouple Amplifier
 - c. Solid-State Relay (to turn griddle on/off), May also be used to turn blender on/off
3. Develop a MATLAB App that displays information from a variety of sensors and signals including:
 - a. Inputs
 - i. Analog Voltage Signals (from instrumentation amplifiers)
 - ii. Digital Input Signals (button pressing, optional)
 - iii. User Inputs (clicks/typing)
 - b. Outputs
 - i. Digital Outputs (Turn griddle on and off (required), turn on/off blender (required), external LED (optional))
 - ii. Turn on/Change Indicators/gauges on MATLAB App
4. Save data from load cells, thermocouples, and timers that may be reviewed after the experiment/process is complete.
 - a. All data should be exported into an excel or csv file with informative headers and must include units (grams, °F, etc.)

- b. Additional information that may be useful for the restaurant to keep track of (total mixing time, total cooking time, pancake weight, date and time cooking began) may also be included.
5. Make decisions on how to design instrumentation and measurement systems to match the application.

Laboratory Methods

You will receive your equipment once your single line, wiring and loop diagrams have been reviewed.

Note: If you'd like to use some other instrumentation/tool to improve your measurement system, please let your instructor/TA know. We will try our best to support your idea with hardware from the lab.

Hardware

- Breadboard
- Jumper Wires
- Terminal Blocks (Q=12)
- Male and Female Barrel Jack Connectors
- LEDs
- RGB LEDs
- Breadboard Pushbuttons

Power Supplies/Converters/Transformers

- LM7805CT Linear Voltage Regulator
- Solid-State Relay (GTZ-1 D4825, 3-32 VDC input, 24-480 VAC 50-60 Hz, 25A output)
- DROK +/- 15V Power Converter
- 5V 2A Power Supply

Sensors/Transducers

- K-Type Thermocouple
- Uxcell 2 kg load cell (Harfington product ID: 1034748)

Signal Conditioning (Amplifiers/Filters)

- K-Type Analog Thermocouple Amplifier
 - Adafruit 1778 Board
- AD 8429 Instrumentation Amplifier

Data Acquisition/Display (Multimeters, DAQ, Oscilloscopes)

- NI USB 6001
- USB cable/USB C adapter
- Computer with MATLAB and NI MAX Installed

Suggested Procedure to complete this project

Day 1 (3/30)-Planning

- Single Line Diagram (1 diagram for entire system)- This may be initially hand drawn but will need to be documented eventually using a CAD program.
- Wiring and Loop Diagrams (1 for force measuring system, 1 for temperature measurement/control, 1 for blender control) (see end of document for templates). These may initially be hand drawn but will need to be documented eventually using a CAD program such as Autodesk Inventor/Autodesk AutoCAD Electrical/NX.
- Averages and acceptable range (XX+/- YY grams) for each ingredient determined with any outlier data points removed. Please present this information within a single table indicating ingredient name, minimum acceptable mass, target mass, and max acceptable mass.
- Pick an appropriate gain and corresponding gain resistor for the load cell's instrumentation amplifier. Explain your reasoning behind the gain selected.
- Identify the relationship between the temperature read and output voltage for the thermocouple amplifier. Create a table that uses this relationship and lists the output voltage at the following temperatures: 70 °F, 150 °F, 250 °F, 375 °F, and 400 °F.

These deliverables will be due on (4/1).

Day 2 & 3 (4/2, 4/6)- Verification and Calibration of Measurement System

Your group must complete the following tasks:

- Verify the wiring & loop diagrams the load cell amplifier, thermocouple amplifier, and NI board circuits.
- Once your system is working. record the voltages in the voltage tables to allow for easier troubleshooting later.
- Read data from both load cells and the thermocouple using the NI Board and MATLAB/NI MAX.
 - Calibrate the load cell/load cell amplifier using known masses
 - Verify the thermocouple/thermocouple amplifier output voltages as a function of temperature using a reference temperature measurement (use Keysight 970A channel 101 and another Type K thermocouple to measure griddle temperature at same time as reading voltage from your group's thermocouple amplifier).
- Create calibration sheets for both the load cell/load cell amplifier and thermocouple/thermocouple amplifier.
- Program the app to be able to "tare" or "zero" the load cell. This functionality will be needed for the user to properly measure the ingredients.

These deliverables are due by the start Day 4.

Day 4 (4/9)- Calibration Review and App Designer Start

By the start of this session, the hardware for your instrument should be documented and verified. Additionally, each group's hardware will be evaluated by the instructor(s) (see Technical Evaluation). Due to the number of groups, your group will get ~10 minutes to show that your groups hardware is working so make sure your equipment is working before the class begins.

This session will be used for your group to use MATLAB's app designer feature [1] to create an application that guides the user on how to make pancakes. Before you begin coding your application, it is critical that you have a plan on how to design the application. A simple way to plan out your program is to develop a flow diagram of the pancake making process from the user's perspective.

- Create a flow diagram of how your group's app will instruct the user on how to make pancakes. Please do not just copy the flow diagram from above, think about what information the user needs to make pancakes and include it in your diagram. It is critical to be detailed so you can properly use the diagram to plan your program. This program will serve as your roadmap to create your code. It is much easier to review and edit your code at the flow diagram stage than to make edits after your code is nearly written. A good resource on creating a flowchart may be found here: <https://www.geeksforgeeks.org/competitive-programming/an-introduction-to-flowcharts/>

Before leaving class, your group should have a clear plan on how to complete the application within the next week.

Deliverables from this day will be due at the end of the evening. The deliverables for this day are:

- A flow diagram detailing the entire pancake making process and what instructions need to be given to the user.
- A comprehensive list of tasks that need to be completed by the end of the project with assigned deadlines.
- Scheduled dates outside of class time to meet to complete the project.

Day 5 (4/13)

This session will be for you to continue to work on your application and to work on your project report. The course instructor will be available for you to ask any questions. At the end of the class session, the task list from last class should be updated to reflect the group's progress.

- Pancake ingredients will be available at this time if you'd like to test your application.

Day 6 (4/16): Preliminary Pancake Application Evaluation

During this class, the instructor/TAs will use your application to make a batch of pancakes after you have given them a 5 minute debrief of how your application works. They will comment on the strengths of the current system and identify any flaws/possible improvements to be made.

The rest of this session will be for you to continue to work on your application and to work on your project report. At the end of the class session, the task list should be updated to reflect your progress.

Day 7- (4/20)

Continue to work on your application and to work on your project report. At the end of the class session, the task list should be updated to reflect your progress.

- Pancake ingredients will be available at this time if you'd like to test your application.

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Day 8 (4/23)

On this day, there will be three/four stations where a Miami faculty or staff member will make your pancakes using your application and measurement system. Please make sure your group is set up and ready to go at the start of class. Once the faculty/staff member is done making the pancakes, clean your cooking utensils and then eat those pancakes! After we test your app and finish eating, the remainder of the time may be used work on the project report.

Section A (4/23)-Assigned Times

Time	Station 1	Station 2	Station 3
8:00-8:30 AM	Group A1	Group A2	Group A3
8:30-9:00 AM	Group A7	Group A8	Group A9
9:00-9:30 AM	Group A4	Group A5	Group A6
9:30 -10:00 AM	Group A10		

Section B (4/23)-Assigned Times

Time	Station 1	Station 2	Station 3
10:00-10:30 AM	Group B1	Group B2	Group B3
10:30-11:00 AM	Group B4	Group B5	Group B6
11:00-11:30 AM	Group B7	Group B8	Group B9
11:30 AM-12:00 PM	Group B10		

Section C (4/23)-Assigned Times

Time	Station 1	Station 2	Station 3
12:05-12:35 PM	Group C1	Group C2	Group C3
12:35-1:05 PM	Group C7	Group C8	Group C9
1:05-1:35 PM	Group C4	Group C5	Group C6
1:35 -2:05 PM	Group C10		

Section D (4/23)-Assigned Times

Time	Station 1	Station 2	Station 3
2:15-2:45 PM	Group D1	Group D2	Group D3
2:45-3:15 PM	Group D7	Group D8	Group D9
3:15-3:45 PM	Group D4	Group D5	Group D6
3:45 -4:15 PM	Group D10		

Suggested Data Analysis Procedure

The type and amount of data analysis you complete will depend on how your group approaches the project.

Project Grading Scheme

Assignment Name	Points	Due Date (R F-Thursday Friday Lab Sections)
Day 1 Deliverables	5	4/1 (R)

Technical Evaluation	90	Document Upload: 4/8 (R), In person evaluation: 4/9 (R),
Day 4 Deliverables	5	4/9 (R)
App Preliminary Evaluation	20	4/16 (R),
App User Evaluation	60	4/23 (R),
Final Report	120	4/29 (R)
Extra Credit (Optional)	30	4/23 (R)

Project Report Deliverables

Throughout this course, you have practiced writing for a target audience in your lab reports/deliverables. For the course project, your group will write a report to the breakfast chain's corporate board explaining why they should adopt your application. This report is meant to serve as formal writing practice. Therefore, please cite any external references sources and proofread your work before submitting. This should be written in paragraph form. Make sure to include axes labels, chart titles, and figure captions in the report. The report should include the following sections.

- 1) Executive Summary: In this section, explain the goals/desired outcomes of the measurement/control system and MATLAB app. Then succinctly explain how the system your group designed achieved that goal.
- 2) Measurement System Design: This section should explain how the measurement system works together and its overall capabilities (sensors, transducers, amplifiers, data acquisition system, etc.) . Detailed information (such as wiring diagrams, loop diagrams, calibration sheets, a bill of materials, and budget) may be referenced here but should be placed within an appendix. These appendices should be detailed enough so the reader could recreate the measurement system.
- 3) Description of Work Flow: This section should provide the reader a description of how each part of pancake manufacturing process may be completed using the app and what information was recorded after each step of the app.
- 4) User Experience: In this section, explain how much better the app was able to improve the pancake manufacturing process. You can compare how each stage of the process was automated/instrumented/modified to improve the process against the original mixing and cooking process that was used. If there are any additional ways the process could be improved, state them and support how these methods would improve the process. **Remember, you're trying to sell this system to the executive board so be convincing of why they should adapt your groups system!**

- 5) Conclusions: Summarize the critical findings from the previous sections. Explain how the app that you developed makes the process more efficient and more consistent than before.
- 6) Appendices: To minimize the body of the report, please use appendices. These may be used to explain in greater detail certain sections of the report. Required appendices include:
 - a. Bill of Materials & Budget: List all the equipment used in your measurement and instrumentation system. Please include the name of the manufacturer, the part number, link to a vendor where the item may be purchased, and the cost of each item. Please give the estimated cost for a single measurement system at the end of this appendix.
 - b. Wiring and Loop Diagrams: Include wiring and loop diagrams for the force measurement, temperature control and blender systems. Please make sure to include all necessary pin labels and wire colors for each device as well as the part # and manufacturer name. You must use a CAD program such as Autodesk Inventor, Autodesk AutoCAD Electrical, or NX to make the diagrams.
 - c. Calibration Sheets: Include calibration files for the thermocouple amplifier and the load cell amplifier. Make sure to include all the critical information used for the calibration process.

1 Word Document & Your Completed App must be submitted.

For help with technical writing, please refer to the MME Writing Guide:

<https://docs.google.com/document/d/1XmCN7NOLGfGhOkX2FGmwAoz2UQzW6zGcNathmhles94/edit?usp=sharing>

Audience for your report

There are two target audiences for this report. The first target audience is the executive board of the restaurant chain who hired you to complete this work. They are not engineers by training and are busy, so they would prefer the body of the report to focus on the bigger picture (how the processes implemented improved the manufacturing process and how much the system will cost to implement). Visual aids such as flow diagrams, images, sketches may help them understand the product you're building. The second target audience is the chief technology officer who would oversee setting up the system you proposed. They need more details on how the system works so they could build these systems and implement them in each restaurant. These details may be put within the appendices of the document.

Please keep the following in mind when writing the project report.

- Include the name of each member of your group.
- Each figure/chart should have a caption underneath it.
- All data and plots should include labeled axes with units of measurement.
- Please reread the project description before submitting your work to ensure you have included all necessary deliverables in your report.

References

- [1] "Develop Apps Using App Designer"[Online]. Available:
https://www.mathworks.com/help/matlab/app-designer.html?s_tid=CRUX_topnav .[Accessed: 15-April-2021]
- [2] "Industry 4.0 - Bosch Rexroth Multi Product Line" [Online]
<https://www.youtube.com/watch?v=DrE0FShBfF4> [Accessed 11-November-2021]
- [3] "POKE YOKE "Mistake Proofing"- Screw Tightening Instruction System" [Online]
https://www.youtube.com/watch?v=hJk7ILvjy_0 [Accessed 30-October-2022]
- [4] "TT-500 Timer" <https://www.kitchenbrains.com/product/fastimer-tt-500/> [Accessed 10/26/2023]

Additional references that may be helpful during this project may be found in previous handouts. Some important ones include:

- Week 3: Load Cells
- Week 6: Thermocouples, TC Amplifiers,
- Week 7: Bang-Bang Control, and Matlab App Designer

Technical Evaluation Review

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1. Criterion #1: Load Cell and Amplifier

27 pts

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Load Cell and Amplifier properly able to measure mass and tare load for expected range of ingredient masses. Calibration Sheet includes all necessary information describing the calibration process.

> Rating Scale: 4
-
- ⋮

2. Criterion #2: Thermocouple and Amplifier

27 pts

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Thermocouple and Thermocouple Amplifier able to properly measure temperature over expected temperature range of the system. Calibration Sheet includes all necessary information describing the calibration process.

> Rating Scale: 4
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3. Criterion #3: Single Line, Wiring & Loop Diagrams

18 pts

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Single Line Diagram: Made using software such as Autodesk Inventor, AutoDesk Electrical, NX. All text is typeset and all lines were created using a computer not stylus or pen/pencil. Shows functional relationships between all sensors, signal conditioners, DAQs, and other hardware within the system

Wiring Diagram: includes all components within measurement and control system, includes manufacturer names, part #s; and was made using software such as Autodesk Inventor, AutoDesk Electrical, NX. All text is typeset and all lines were created using a computer not stylus or pen/pencil.

Loop Diagram includes all components within measurement and control system, includes manufacturer names, part #s; and was made using software such as Autodesk Inventor, AutoDesk Electrical, NX. All text is typeset and all lines were created using a computer not stylus or pen/pencil. Should include date, title, instrument tags, input/output calibrations, and notes.

All diagrams must also construct a working system.

> Rating Scale: 4
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- ⋮

4. Criterion #4: Voltage Table

9 pts

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Voltage table a list of pin #s/Terminal blocks for each component, whether that terminal block/pin is connected to another pin/device, and the voltage of that pin/device should be at when the device is powered.

> Rating Scale: 4
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- ⋮

5. Criterion 5: Analog and/or Digital Filter

9 pts

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An Analog and/or Digital Filter must be used to improve the performance of the load cell. The group must provide quantitative evidence that the implementation of the filter improved the performance of the load measuring system.

Preliminary App Evaluation

1. Measuring and Mixing Ingredients

5 pts

-Excellent: Very obvious on what ingredient/how much is to be used and clear indication of when the ingredient was within the desired limits; -Very Good: Easy to use during measuring ingredient phase. Some challenge to determine if step was completed properly.; -Good: User able to properly complete measuring and mixing process but with great difficulty/took a very long time. ; -Poor: User unable to properly complete measuring and mixing process based on instructions provided by user.

> Rating Scale: 5

2. Ease of cooking pancake.

5 pts

-Excellent: Very obvious on whether griddle was at proper temperature, how much batter to ladle onto the griddle and timer clearly notified user when to flip and remove pancakes. -Very Good: Easy to use during cooking phase. Some challenge to determine if griddle was in correct range or when action (flip/remove pancake) was needed.; -Good: User able to properly complete cooking process but with great difficulty/took a very long time. ; -Poor: User unable to properly complete cooking phase based on instructions provided by user.

> Rating Scale: 5

3. Timer/Video/Audio Cues

5 pts

Timer is able to be reset and provides indication that timer has ended, Software Indicator (Lamp/Checkbox etc) or Hardware (LED) that indicates if T or mass is in range. Audio cue (like a beep or other sound) used to get users attention

> Rating Scale: 5

4. Data Storage, Heat Control System, Blender Control System

5 pts

Mass and Temp Data stored in excel, csv, matlab with appropriate headers, identifiers, and units, Use of a solid state relay to control temperature of griddle. Group must prove that the system can keep system with +/-5 °F of setpoint for 5 mins, System can blend ingredients for a specific amount of time then stop automatically.

> Rating Scale: 5

Qualitative App Evaluation Review (1x guest, 1x TA)- -Fill out both Front & Back

Students' Last Names & Section: _____ Evaluator Name: _____

Evaluation Instructions

1. This is not a grading exercise (5 ≠A, 1≠F), please be honest and straightforward on your feedback.
2. Fill in the back of the evaluation sheet to provide students with qualitative feedback on the strengths and possible improvements they could make to their app.

Criterion #1: Ease of Measuring Ingredients and App Navigation

1a. Instructions given by the group and application's format effectively helped you navigate the pancake making process.

Strongly Disagree (1)	Somewhat Disagree (2)	About the same (3)	Somewhat agree (4)	Strongly Agree (5)
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1b. Application and hardware made it easy to measure the correct amount of each ingredient

Strongly Disagree (1)	Somewhat Disagree (2)	About the same (3)	Somewhat agree (4)	Strongly Agree (5)
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1c. The application had obvious and clear indicator(s) (Lamp on Screen, LED, buzzer) when the ingredient was within the appropriate mass range.

Strongly Disagree (1)	Somewhat Disagree (2)	About the same (3)	Somewhat agree (4)	Strongly Agree (5)
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Comments for Criterion 1:

Criterion #2: Ease of Mixing Ingredients/ Cooking Pancake

2a. The application clearly explained how to mix the ingredients and for how long.

Strongly Disagree (1)	Somewhat Disagree (2)	Sort of (3)	Somewhat agree (4)	Strongly Agree (5)
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2b. It was obvious that the griddle was at the appropriate temperature before and during cooking.

Strongly Disagree (1)	Somewhat Disagree (2)	Sort of (3)	Somewhat agree (4)	Strongly Agree (5)
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2c. The timer was easy to start/reset and was clear when it was time to flip/remove pancakes.

Strongly Disagree (1)	Somewhat Disagree (2)	Sort of (3)	Somewhat agree (4)	Strongly Agree (5)
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Comments for Criterion 2:

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Strengths of the Application: (Please use this section to describe what parts of the application you enjoyed using the most or any other compliments you'd like to give to the group).

Possible Improvements to the Application: [Please use this section to explain what parts of the application were difficult to use and possible ways they could be improved]

Extra Credit (30 points)

NOTE: If you're in a group of 3 people, you must do at least 2/3 of the extra credits. These will be used as part of your total user evaluation grade. You may do additional tasks to earn additional extra credit.

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1. Mouse/Keyboard Free Operation

10 pts

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Once the user opens the app, the program automatically guides them through the pancake making process without them having to click the mouse or press any keys on the keyboard. Successful operation of the app by the guest user and instructor will be used to grade the effectiveness of this part of the app.

> Rating Scale: 5
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- ⋮

2. User Selected Recipe

10 pts

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Once the user opens the app, the program will allow them to upload their own pancake/crepe recipe file (Swedish Pancake, Cameroonian Pancakes, Crepes). The pancake/crepe recipe file should include all ingredient masses, mixing time, cooking time, cooking temperature, and other instructions the user would need to have.

> Rating Scale: 5
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- ⋮

3. Automatic Load Cell Calibration

10 pts

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The user of the application can set the calibration constant used in the program by placing a known weight onto the scale. This process should automatically adjust the calibration constants within the user's program.

> Rating Scale: 5

Wiring Diagram Templates & Tables

Use these tables when setting up your circuits to help troubleshoot connections using your multimeter. Once you have your system set up and running, record the voltages here. The next time you setup your system, refer to these tables to verify everything is functioning as expected.

Amplifiers

InAmp

AD8429 Instrumentation Amplifier			
Pin #	Function	Voltage rel. to GND	Connected or not?
1			
2			
3			
4			
5			
6			
7			
8			

TT

Adafruit 1778 (AD8495)-Type K Thermocouple Amplifier			
TB #	Function	Expected Voltage relative to Ground	Connected or not?
V+			
GND			
Out			
GND			
Ylw (+)			
Red (-)			

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Power Converters/Regulators

PCM

DROK Power Converter (+/-15 V Supply)			
Pin Label	Function	Expected Voltage relative to Ground	Connected or not?
V _{in}			
GND			
EN			
V _{o+}			
GND			
V _{o-}			

LVR with 10 kOhm potentiometer on pins 3, 4, &5 (note pin 5 is jumped to pin 2)

LM7805CT Linear Voltage Regulator			
Terminal	Function	Expected Voltage relative to Ground	Connected or not?
1			
2			
3			
4			
5			

This table will depend on the group's design

TB

Terminal Blocks			
TB #	Function	Expected Voltage relative to Ground	Connected or not?
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			

DAQ and Control

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DAQ (terminals may vary depend on design)

NI 6001 DAQ			
Terminal	Function	Expected Voltage relative to Ground	Connected or not?
AI0+			
AI0-			
AI1+			
AI1-			
P0.0			
P0.1			
Dig GND			

SSR 1

Solid-State Relay (GTZ-1 D4825, 3-32 VDC input, 24-480 VAC 50-60 Hz, 25A output)			
Terminal	Function	Expected Voltage relative to Ground	Connected or not?
1			
2			
3			
4			

SSR 2

Solid-State Relay (GTZ-1 D4825, 3-32 VDC input, 24-480 VAC 50-60 Hz, 25A output)			
Terminal	Function	Expected Voltage relative to Ground	Connected or not?
1			
2			
3			
4			

Single Line Diagram

Legend

-Solid lines (Power)

-Dash line (Analog Signal)

-Dotted line (Digital signals)

